13. Coding & Solution

Date	30 June 2026
Team ID	SWTID-2026-8135
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Summary

Repository Link / URL	https://github.com/deepthi3535/Edu_Engine_Google
Programming Language(s)	Python 3.10, JavaScript, HTML, CSS
Framework(s) Used	FastAPI, Uvicorn, Jinja2
Key Features Implemented	AI Q&A chat; leveled concept explanation (local T5 + Gemini fallback); MCQ quiz generation; text/notes summarization; personalized learning roadmap; JSON-file query/response logging; demo/mock mode; Dockerized deployment
Pending / Incomplete Features	User authentication and role-based access; migration from JSON files to a relational database; usage analytics dashboard; consolidation of duplicate router/direct API endpoints
Setup / Run Instructions	1) pip install -r requirements.txt 2) Copy .env.example to .env and set GEMINI_API_KEY 3) uvicorn main:app --reload 4) Open http://127.0.0.1:8000 Docker: docker build -t edugenie . then docker run -p 8000:8000 edugenie

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes (verified via tests/test_api.py, all routes return 200/422 as expected)
6	Code is committed to a version control repository	Yes (GitHub)

Additional Notes / Comments

The application was verified locally in demo/mock mode (no GEMINI_API_KEY configured) using FastAPI's TestClient — all page routes, the five router-based endpoints and the five direct endpoints returned successful responses. CORS is currently open to all origins for local testing and should be restricted before any public deployment.